

Contribution submission to the conference Dresden 2026

Normal Incidence X-ray Standing Wave (NIXSW) technique of Quinacridone on the Ag(100) surface — ●LUCAS HIRSCHFELD
— Clausius Institute for Physical and Theoretical Chemistry, University of Bonn, Germany

Vacuum deposited films of quinacridone (QA, 5,12-dihydro-quinol[2,3-*b*]acridine-7,14-dione) on the Ag(100) surface have been investigated with focus on their ordering via hydrogen bonds and the change of the adsorbed molecule during the phase transition using Normal Incidence X-ray Standing Wave (NIXSW) technique in combination with X-ray Photoelectron Spectroscopy (XPS) performed at the Diamond Light Source in UK.

QA on the Ag(100) surface forms a point-on-line structure at room temperature (α -phase), which transforms into an commensurate and thermodynamically favored β -phase upon annealing to 500 K [1]. The NIXSW and XPS measurements revealed that the phase transition involves a partly deprotonation of the amine group and a decrease of the adsorption height for the complete molecule and especially for the deprotonated nitrogen atoms.

[1] Journal of Physical Chemistry C 124.45 (2020), 2486124873.

Part: O
Type: Poster
Topic: Organic molecules on inorganic substrates:
electronic, optical and other properties
Keywords: NIXSW; XPS; Quinacridone; Ag(100)
Email: hirschfeld@pc.uni-bonn.de